DS 501 · INTRODUCTION TO DATA SCIENCE

Week 05

Data Cleaning, Wrangling and EDA

How to find and fix data quality problems, reshape and combine data with pandas, and explore it before modelling.

THIS WEEK

- **5-1** Data Quality and Cleaning — 45 minutes
- **5-2** Data Wrangling with pandas — 45 minutes
- **5-3** Exploratory Data Analysis — 45 minutes
- **P5** Practice: cleaning and EDA on a messy Titanic — 60 minutes · Python notebook

International Balkan University · [Instructor name] · [Semester]

Recap of Week 4

Three ideas from Databases and SQL that this week builds on:

Keys identify rows

A **primary key** identifies each row; a foreign key points to a row in another table. This week, keys define duplicates and decide whether a merge is safe.

SQL operations

SELECT, WHERE, JOIN and GROUP BY answer most questions. This week you meet their pandas equivalents: filtering, **merge** and **groupby**.

Many sources

Warehouses, lakes, NoSQL stores and APIs each follow their own codes and formats. Combining them creates the **quality problems** you learn to find.

Keep your Week 4 SQL notes open: almost every pandas operation this week has an SQL twin.

DS 501 · WEEK 5 · LESSON

5-1

Data Quality and Cleaning

Lesson 1 of 3

In this lesson

01 What data quality means, and how to report it

02 Why values are missing, and what to do about it

03 Outliers, duplicates and inconsistent codes

04 A repeatable cleaning workflow without leakage

45 minutes

Builds on weeks 2–4

Objectives

By the end of this lesson, you will be able to:

01 **Describe** the six dimensions of data quality

Checked in: Quiz 5 · question 1

02 **Distinguish** the mechanisms MCAR, MAR and MNAR

Checked in: Quiz 5 · question 2

03 **Decide** on a defensible strategy for missing values

Checked in: Quiz 5 · question 3

04 **Decide** how to treat outliers and duplicates

Checked in: Quiz 5 · question 4

05 **Design** a repeatable cleaning workflow without leakage

Checked in: Quiz 5 · questions 5 and 16

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Objective 01

Case: “Is This Data Ready?”

THE SITUATION

A hospital sends a data science team 10,000 patient records. A first look shows ages of 0, −1 and 999, sex coded as M, F, male and 1, and weights blank for one patient in five.

> “We are in a hurry. Can you start modelling today?”

The figures are illustrative.

YOUR TASK · PAIRS · 5 MIN

1. Which of these problems could you fix immediately, and which need more information?
2. What could an age of 0 mean? And an age of 999?
3. What would you ask the data owner **before** filling the blank weights?

Objective 01

Six Dimensions of Data Quality

Data quality means **fitness for purpose**. Check each dimension for every important variable:

Dimension	Question to ask	Example of a problem
Completeness	Are required values present?	One patient weight in five is blank
Validity	Do values follow the allowed format and range?	An age of 999; a date of 31 February
Consistency	Is the same thing recorded the same way?	Sex coded M, Male and 1 in different files
Uniqueness	Is each real entity recorded once?	One admission entered by two departments
Accuracy	Does the value match reality?	An old address after a customer moved
Timeliness	Is the data recent enough for the question?	A price list last updated two years ago

In practice

Five of the six dimensions can be checked with a few lines of pandas.

Critical point: accuracy is the only dimension you cannot check with code alone. It needs an outside source of truth.

Objective 01

The Data Quality Report

Record each problem, its dimension and a decision before you change anything. Example: the messy Titanic file.

Variable	Problem found	Dimension	Decision
sex	7 spellings, e.g. “M”, “ male”, “Female ”	Consistency	Map to male / female
fare	Text with “GBP” and decimal commas	Validity	Convert to a number
age	4 values below 0 or above 100	Validity	Set to missing
age	20.3% missing	Completeness	Impute by group; flag
passenger_id	12 repeated rows	Uniqueness	Drop the repeats
deck	77.2% missing	Completeness	Drop the column

← **Read the last column first.** Every problem ends in a decision that a reviewer can check.

In practice

Your data audit this week uses the same four columns for your own project dataset.

Why Is It Missing? Three Mechanisms

Rubin (1976) classified missing data by what the missingness depends on:

MCAR

Missing completely at random: unrelated to any variable, observed or not.

Example: a blood sample is lost because a tube broke.

Effect: fewer data, but no bias.

MAR

Missing at random: depends only on **observed** variables.

Example: older patients skip an online form more often; age is recorded.

Effect: bias, which observed variables can correct.

MNAR

Missing not at random: depends on the missing value itself.

Example: people with very high incomes refuse to report income.

Effect: bias that the data alone cannot correct.

Common mistake: reading “missing at random” as “missing for no reason”. MAR means random **once you account for** other observed variables.

Objective 02

Ask why a value is missing before you decide what to put in its place.

Deleting or filling missing values is never neutral. The right choice depends on what caused the gaps, and that knowledge comes from how the data was collected, not from a pandas function.

- Rule of thumb: compare the rows with and without the value. If they differ, the data are not MCAR.

Objective 03

Strategies for Missing Values

Strategy	How it works	Safe when	Risk
Listwise deletion	Drop rows with a missing value	MCAR and few rows lost	Bias under MAR or MNAR
Simple imputation	Fill with the mean, median or mode	Very few values missing	Shrinks the spread
Group imputation	Fill with the median of a group	MAR given the groups	Groups must be chosen well
Model-based imputation	Predict from other variables (k-NN, MICE)	MAR with good predictors	Complex; fit on training data
Missing indicator	Add a 0/1 column “was missing”	Missingness is informative	Not a fix on its own

← **Practice 5 uses this row**, plus a missing indicator, for Titanic ages.

Key point: an imputed value is an estimate. Keep a flag, and check that your conclusions hold without it.

Missing Values in Practice: Titanic Ages

Share of passengers without a recorded age, after cleaning (891 passengers):

Passenger class	Age missing
First class	14%
Second class	6%
Third class	28%
All passengers	20.3%

⚠ **Common mistake:** dropping the rows without age. You would remove mostly third-class passengers, who died more often.

← **Read the third-class row.** Missingness depends on class, so age is not MCAR.

- If age were MCAR, the three class shares would be roughly equal; they differ by a factor of almost five.
- Complete cases contain too few third-class passengers, so their survival rate is too high.

Outliers: Error or Discovery?

- An **outlier** lies far from most other observations. It may be an error, or the most important value in the data.
- Rules help to find them: the **IQR rule** (beyond 1.5 × IQR from the quartiles), z-scores, domain limits and plots.
- An **impossible** value is corrected from the source or set to missing; a **possible but extreme** value is kept.
- In fraud, machine failures and rare diseases, the outlier may be exactly what you are looking for.

Messy Titanic ages (723 values)

Mean 32.48 · median 28.00

Standard deviation 53.63

Minimum −1 · maximum 999

Quartiles 20 and 38 years

Why? The IQR fences, −7 and 65, miss the age of −1 and flag real passengers over 65.

Duplicates: Same Row or Same Entity?

Before you drop a row, decide what makes two rows the **same**.

True duplicate

Two rows share the same **identifier**.

Messy Titanic: **12** passenger_id values repeat; dropping them leaves 891 rows.

Decision: keep one copy.

Look-alike rows

Different people with identical attributes.

Clean Titanic: **107** rows match another row on every column except the ID.

Decision: keep them all.

Fuzzy duplicate

The same entity, spelled differently.

“Ana Petrovska, Skopje” and “A. Petrovska, SKOPJE” (fictional).

Decision: standardise, then match.

Critical point: define a duplicate by a true identifier, never by identical values alone.

Objective 04

Fixing Codes and Types in Code

```python
# Python · pandas
s = df["sex"].str.strip().str.lower()
df["sex"] = s.replace({"m": "male", "f": "female"})
assert set(df["sex"]) == {"male", "female"}
f = df["fare"].str.replace(" GBP", "", regex=False)
f = f.str.replace(",", ".", regex=False)
df["fare"] = pd.to_numeric(f, errors="coerce")
```

1–2 Strip spaces, lower the case, then map short codes with a dictionary.

← **3 · Critical:** an assertion stops the notebook if a new spelling ever appears.

4–5 Remove the currency text and replace the decimal comma.

6 Convert to numbers; anything left unconvertible becomes NaN.

⚠ **Common mistake:** fixing values by hand in a spreadsheet. Nobody can rerun or review a manual edit.

A Repeatable Cleaning Workflow

01

Keep the raw data. Never overwrite the original file; work on a copy.

02

Profile: types, missing values, ranges, duplicates and codes, recorded in the quality report.

03

Decide for each problem: fix, flag, accept or ask the data owner.

04

Clean in code, so that every step can be rerun on the next file.

05

Separate fixed rules from learned values: learn medians and thresholds on training data only.

06

Validate with assertions: allowed values, ranges, unique keys and row counts.

Key point: a rule such as “999 means missing” is safe everywhere; a median used for imputation must come from the training data.

Objective 01

Exercise: Audit a Student Records Export

THE EXPORT (FICTIONAL)

"We need a clean list of active students by Friday."

1. Programme written as "Computer Engineering", "CE" and "comp. eng."
2. Two students born on 31/02/2001.
3. Phone number blank for 35% of students.
4. Student ID 21045 appears twice, with different e-mails.
5. A grade average of 11.4 on the 5–10 scale.

YOUR TASK · PAIRS · 6 MIN

1. Name the quality dimension of each problem.
2. For problem 3, which mechanism is plausible, and how would you check it?
3. Choose a decision for each problem: fix, flag, accept or ask.

Check Your Understanding

In a survey, income is blank for 30% of respondents, and the blanks come mostly from people with the highest incomes. Which statement is correct?

A The data are MCAR, so dropping the blank rows gives an unbiased average.

B The data are MNAR, so the average of the reported incomes is too low.

C The data are MAR, because 30% is too many values to be missing at random.

D Filling the blanks with the mean of the reported incomes removes the bias.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Data quality
The degree to which data is fit for its intended use.

Data quality report
A record of each problem found, its dimension and the decision taken.

MCAR
Missingness unrelated to any variable, observed or unobserved.

MAR
Missingness that depends only on other observed variables.

MNAR
Missingness that depends on the missing value itself.

Imputation
Replacing missing values with estimated values.

Outlier
An observation that lies far from most other observations.

Data leakage
Information from test data or the future entering model training.

Summary

In this lesson, you learned how to:

- **O1** Describe the six dimensions of data quality
- **O2** Distinguish the mechanisms MCAR, MAR and MNAR
- **O3** Decide on a defensible strategy for missing values
- **O4** Decide how to treat outliers and duplicates
- **O5** Design a repeatable cleaning workflow without leakage

Key takeaway: find and document every problem first; then fix it in code, with a decision someone can review.

Next lesson

5-2 Data Wrangling with pandas

DS 501 · WEEK 5 · LESSON

5-2

Data Wrangling with pandas

Lesson 2 of 3

In this lesson

01 What wrangling is, and how it differs from cleaning

02 Selecting, filtering and reshaping tables

03 Combining tables safely: concat and merge

04 Summaries, features and readable pipelines

45 minutes

Builds on 4-2 and 5-1

Objectives

By the end of this lesson, you will be able to:

01 **Explain** what data wrangling is and where it fits in CRISP-DM

Checked in: Quiz 5 · question 6

02 **Apply** selection, filtering and reshaping in pandas

Checked in: Quiz 5 · question 7

03 **Apply** concatenation and merges, and validate join keys

Checked in: Quiz 5 · question 8

04 **Apply** groupby, aggregation, transform and pivot tables

Checked in: Quiz 5 · question 9

05 **Design** new features and a readable, rerunnable wrangling pipeline

Checked in: Quiz 5 · question 10

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Objective 01

Case: One Table for the Model

THE SITUATION

A Skopje online shop wants to predict which customers will stop buying. Its data:

- Sales: one CSV file per month, 24 files.
- Customers: an export from the CRM system.
- Products: an Excel sheet kept by the buyers.

"The model needs one row per customer. How long will that take?"

YOUR TASK · PAIRS · 5 MIN

1. List the steps from the files to one row per customer, in order.
2. What is the **unit of analysis** of each source, and of the final table?
3. At which step could rows be lost or counted twice?

Objective 01

What Is Data Wrangling?

- **Data wrangling** transforms raw or cleaned data into the structure that an analysis or model needs.
- It selects, reshapes, combines and aggregates data to the **unit of analysis**, and constructs new variables.
- In CRISP-DM it belongs to **data preparation**, together with cleaning.
- pandas suits files and exploration; SQL suits large tables that stay in a database.

CRISP-DM preparation tasks

- Select data
- Clean data · lesson 5-1
- Construct data
- Integrate data
- Format data

Key point: cleaning fixes what is **wrong**; wrangling changes the **shape** to fit the question.

Objective 02

Selecting, Filtering and Sorting

```python
# Python · pandas · Titanic
df[["sex", "age", "fare"]]
df[(df["pclass"] == 1) & (df["age"] < 30)]
df.query("pclass == 1 and age < 30")
df.loc[df["survived"] == 1, "fare"]
df.sort_values(["pclass", "fare"], ascending=[True, False])
df.loc[bad_age, "age"] = np.nan
```

1 A list of names selects columns.

2–3 Two ways to filter rows: a boolean mask, or query.

4–5 .loc selects rows and columns by label; sort by two keys.

← **6 · Critical:** change values through **.loc**. A chained df[mask]["age"] = … does not change df.

</> **In practice**

.iloc selects by position instead: df.iloc[:5, :3] gives the first five rows and three columns.

Objective 02

Reshaping: Wide and Long

Wide: one column per week

student	quiz_w1	quiz_w2	quiz_w3
S01	8	7	9
S02	6	5	NaN

```
long = wide.melt(id_vars="student",
  var_name="week", value_name="score")
```

.melt()

.pivot()

Long (tidy): one row per student and week

student	week	score
S01	quiz_w1	8
S01	quiz_w2	7
S01	quiz_w3	9
S02	quiz_w1	6
S02	quiz_w2	5
S02	quiz_w3	NaN

Note: the empty cell becomes its own row with NaN. Drop it or keep it on purpose.

- **Tidy data** (Wickham, 2014): each variable is a column and each observation is a row.
- Long tables suit groupby, plotting and models; wide tables suit people and one-row-per-entity models.

Objective 03

Combining Tables: concat and merge

Operation	pandas	Keeps	Use it to
Concatenate	pd.concat([jan, feb])	All rows of every table	Stack files with the same columns
Inner merge	how="inner"	Only keys found in both	Study matched records only
Left merge	how="left"	Every left row; NaN if no match	Add attributes from a lookup
Outer merge	how="outer"	All keys from both tables	Find records that do not match

In practice

indicator=True adds a column saying whether each row matched in left only, right only or both.

Read this row first. A left merge keeps every observation, so nothing disappears silently.

Objective 03

The Repeated-Key Trap and validate=

orders (3 rows)

order	contract	amount
1	Monthly	70
2	Yearly	85
3	Monthly	60

lookup: key repeated!

contract	label
Monthly	flexible
Yearly	1 year
Monthly	monthly

merged: 5 rows instead of 3

order	contract	amount	label
1	Monthly	70	flexible
1	Monthly	70	monthly
3	Monthly	60	flexible
3	Monthly	60	monthly
2	Yearly	85	1 year

Critical point: no error, no warning; total revenue rises from 215 to 345.

- If the lookup key is not unique, every matching row is **duplicated**.
- Pass **validate="many_to_one"** so pandas raises an error instead.

Objective 03

A merge that runs without an error can still give you the wrong table.

Wrangling mistakes rarely crash. They lose rows, duplicate rows or add missing values quietly, and the wrong number travels into every chart and model that follows.

Rule of thumb: after every merge, filter or concat, check the row count, the key and the new missing values.

Objective 04

Split, Apply, Combine with groupby

```python
# Python · pandas · Practice 5
df.groupby("sex")["survived"].mean().round(3)
pd.crosstab(df["pclass"], df["sex"], values=df["survived"],
            aggfunc="mean").round(3)
df.groupby(["pclass", "sex"])["age"].transform(
    lambda s: s.fillna(s.median()))
```

1 One value per group: survival 0.729 for women, 0.193 for men.

2–3 A table of survival rates for two grouping variables.

4–5 · Critical: transform returns one value per **row**, not per group, so it can fill a column.

</> In practice

Several named summaries at once: .agg(n=("age", "size"), mean_age=("age", "mean")).

Objective 04

Reading a Cross-Tabulation

Survival rate by class and sex, Practice 5 (891 passengers):

Class	Female	Male
1st	0.968	0.374
2nd	0.909	0.159
3rd	0.483	0.140
All classes	0.729	0.193

⚠ **Common mistake:** reporting rates without counts. Small groups give unstable rates, so show n beside every rate.

← **Read the third-class row.** Women still did far better than men, but half as well as women in first class.

Constructing Features

A **feature** expresses domain knowledge more directly than the raw columns:

Combinations

family_size = sibsp + parch + 1: the passenger plus relatives on board.

Ratios

fare_per_person = fare / family_size; price per square metre; debt to income.

Bins

Age bands or income groups with pd.cut and pd.qcut; is_child = age < 16.

Dates

Weekday, month or days since the last purchase, from the .dt accessor.

Text

Length, a keyword, or a title such as “Mrs” extracted with .str.extract.

History

Number of orders in the past year, or the average of the last three bills.

⚠ **Common mistake:** building a feature from information that is not known at prediction time. That is leakage.

Objective 05

Wrangling as a Readable Pipeline

```python
# Python · method chaining · illustrative sales data
customers = (sales
    .merge(products, on="product_id", how="left", validate="m:1")
    .assign(revenue=lambda d: d["qty"] * d["price"])
    .groupby("customer_id")
    .agg(n_orders=("order_id", "nunique"), total=("revenue", "sum"))
    .reset_index())
```

← **2 · Critical:** the check sits inside the pipeline, so it runs every time.

3 Add a column with assign.

4–5 One row per customer, two named summaries.

6 Turn the group key back into a column.

</> **In practice**

Wrap the chain in a function, then **assert customers["customer_id"].is_unique**.

Objective 03

Exercise: Which Operation?

Tick the main pandas operation for each task. Work in pairs for 3 minutes.

Task	concat	merge	groupby	melt
Stack twelve monthly sales files into one table				
Add the product category to each sale				
Compute total revenue per customer				
Turn columns q1, q2, q3 and q4 into rows				
Compute average delivery time per region				

Why? Name the unit of analysis before and after each task; it tells you the operation.

In practice

The SQL twins from week 4: UNION ALL stacks, JOIN merges, GROUP BY groups. SQL has no simple melt.

Check Your Understanding

You add a region name to 10,000 sales rows with a left merge on region_code. The result has 10,450 rows. What is the most likely cause?

A The left merge dropped the sales whose region_code had no match.

B Some region_code values appear more than once in the region table.

C Sales with a missing region_code are turned into extra NaN rows.

D A left merge always adds rows; an inner merge would have been correct.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Data wrangling
Transforming data into the structure an analysis or model needs.

Tidy data
Each variable is a column, each observation a row.

Merge (join)
Combining tables side by side by matching key values.

Split–apply–combine
Split data into groups, apply a function, combine the results.

Unit of analysis
The entity that one row of the final table describes.

Concatenation
Stacking tables with the same columns, one below the other.

validate=
A merge argument that checks the expected key relationship.

Feature construction
Creating new variables that express domain knowledge.

Summary

In this lesson, you learned how to:

- **O1** Explain what data wrangling is and where it fits in CRISP-DM
- **O2** Apply selection, filtering and reshaping in pandas
- **O3** Apply concatenation and merges, and validate join keys
- **O4** Apply groupby, aggregation, transform and pivot tables
- **O5** Design new features and a readable, rerunnable wrangling pipeline

Key takeaway: wrangle to one row per unit of analysis, and check the row count after every step.

Next lesson
5-3 Exploratory Data Analysis

DS 501 · WEEK 5 · LESSON

5-3

Exploratory Data Analysis

Lesson 3 of 3

In this lesson

01 What exploratory analysis is for, and a workflow

02 One variable at a time, then pairs of variables

03 Why summaries are not enough: always plot

04 Confounding, Simpson's paradox and the EDA log

45 minutes

Builds on 5-1 and 5-2

Objectives

By the end of this lesson, you will be able to:

01 **Explain** the purpose of EDA versus confirmatory analysis

Checked in: Quiz 5 · question 11

02 **Decide** the summary and chart for one or two variables

Checked in: Quiz 5 · question 12

03 **Interpret** a correlation, and why summaries are not enough

Checked in: Quiz 5 · question 13

04 **Identify** confounding and Simpson's paradox

Checked in: Quiz 5 · question 14

05 **Design** an EDA log entry that records a hypothesis

Checked in: Quiz 5 · question 15

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Case: Does the App Make Customers Spend More?

THE SITUATION

A retail chain finds that customers who use its mobile app spend much more per year than customers who do not. The example is hypothetical.

> “The numbers are clear. Let's give every customer a discount to install the app.”

YOUR TASK · PAIRS · 5 MIN

1. Give two explanations for the pattern in which the app does **not** cause the extra spending.
2. Which data would you explore to tell the explanations apart?
3. What would convince you that the app really causes the extra spending?

Objective 01

Exploratory Versus Confirmatory Analysis

Tukey (1977) compared exploratory analysis to detective work: finding clues before anything is proved.

Exploratory analysis

Goal: find patterns, problems and surprises.

Questions: open, and they change as you look.

Methods: many plots, summaries and group comparisons.

Output: hypotheses and new questions.

Confirmatory analysis

Goal: test a claim with known error rates.

Questions: fixed in advance.

Methods: a planned test or model.

Output: evidence for or against the claim · week 7.

Key point: a pattern found by exploring the data cannot be confirmed with the **same** data.

Objective 01

A Systematic EDA Workflow

01

Write down the questions and the unit of analysis before the first plot.

02

One variable at a time: distribution, centre, spread, missing and odd values.

03

Two variables: each predictor against the target, then important pairs.

04

Within groups: repeat key comparisons inside each level of a likely confounder.

05

Loop back to cleaning whenever a plot reveals a data problem.

06

Record each finding in the EDA log, with its number and the question it raises.

In practice

Practice 5 follows this order: sex and class first, then age by survival, then port within class.

Univariate Analysis: One Variable at a Time

Summaries of two features from Practice 5 (891 passengers):

Statistic	fare_per_person	family_size
Mean	19.92	1.90
Median	8.30	1
Standard deviation	35.84	1.61
Minimum	0.00	1
First and third quartile	7.25 and 23.67	1 and 2
Maximum	512.33	11

⚠ **Common mistake:** describing a skewed variable by its mean alone. Report the median and plot a histogram.

← **Read the first two rows.** A mean more than twice the median means strong right skew.

Bivariate Analysis: Choosing a Method

The types of the two variables decide the summary and the chart:

Variables	Summary	Chart	pandas
Numeric × numeric	Pearson or Spearman correlation	Scatter plot	df[["x", "y"]].corr()
Categorical × numeric	Median or mean per group	Box plot per group	groupby().agg()
Categorical × categorical	Row proportions	Grouped bar chart	pd.crosstab(normalize=…)
Time × numeric	Rolling mean, change	Line chart	resample(), rolling()

In practice

seaborn draws all four: scatterplot, boxplot, countplot and lineplot.

Note: for a 0/1 outcome such as survived, the mean per group **is** the rate per group.

- Show the **count** beside every group summary, so that small groups are not over-interpreted.
- Plot first, then compute the summary that the plot shows is appropriate.

Objective 03

Anscombe's Quartet: Always Plot the Data

- **Pearson's r** measures only how closely points follow a **straight line**, from −1 to +1.
- I is a noisy line; II a curve; III a line with one outlier; IV one point makes the whole trend.

Objective 03

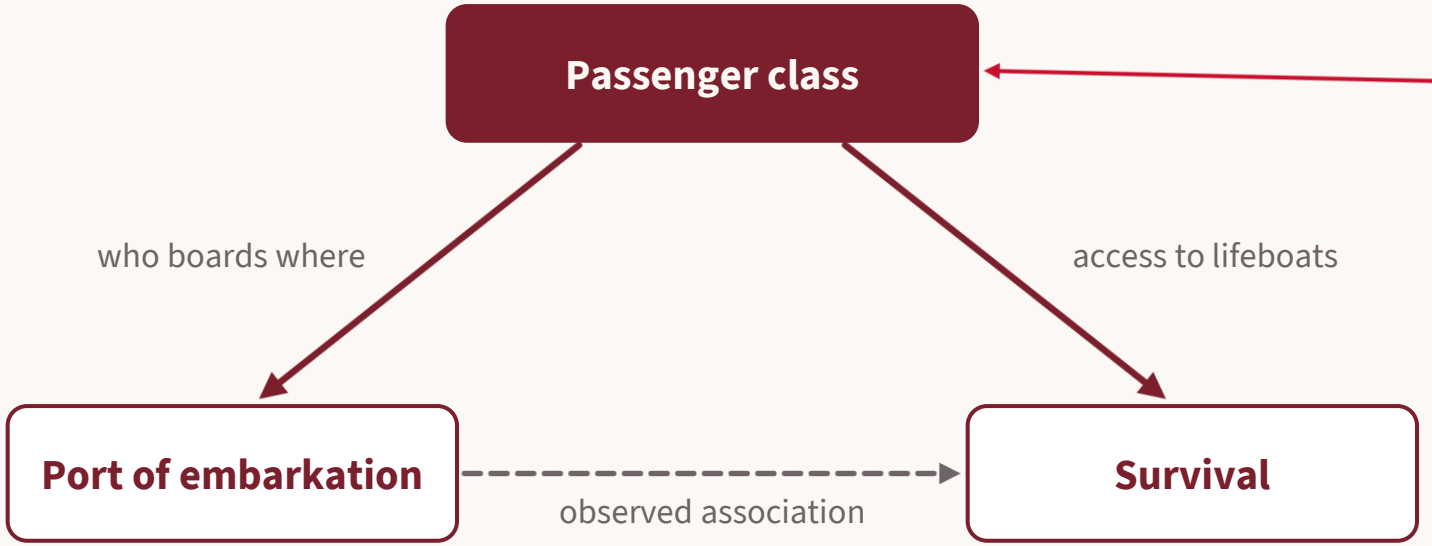
Identical summaries can hide completely different data. Always plot.

A mean, a standard deviation or a correlation compresses many values into one number. The plot shows what the number has thrown away: curves, clusters, outliers and impossible values.

Rule of thumb: for every summary you report, look at one plot that could prove it misleading.

Objective 04

Confounding: A Third Variable



Critical point: the dark box causes both. Port and survival can be linked with no direct effect at all.

Same shape: customer loyalty drives both app use and spending.

- A **confounder** affects both the explanatory variable and the outcome.
- Control for it by comparing **within** its levels, or by including it in a model.

Objective 04

Worked Example: Did Cherbourg Save Lives?

1 · Survival by port

Port	Survived
Cherbourg	55.4%
Queenstown	39.0%
Southampton	33.7%

2 · Class mix by port (row %)

Port	1st	2nd	3rd
Cherbourg	51%	10%	39%
Queenstown	3%	4%	94%
Southampton	20%	25%	55%

3 · Survival within class

Class	Cherb.	Queens.	South.
1st	69%	50%	58%
2nd	53%	67%	46%
3rd	38%	38%	19%

Critical point: half of the Cherbourg passengers travelled first class.

- Within each class, the port gap is smaller and not consistent: in 2nd class, Queenstown is highest.
- Class explains much of the gap; some groups are tiny, such as first-class Queenstown passengers.

Simpson's Paradox

- **Simpson's paradox:** a trend in combined data weakens, disappears or reverses when the data are split into groups.
- It happens when a confounder is distributed **unevenly** across the groups being compared.
- The grouped analysis usually answers the causal question, but not if the grouping variable is itself an effect.

Berkeley admissions, autumn 1973

Overall, about **44%** of men and **35%** of women were admitted.

In four of the six largest departments, women were admitted at a higher rate.

Women applied more often to departments that admitted few applicants of either sex.

Why? The confounder is the department: Bickel, Hammel & O'Connell (1975).

Recording Findings: The EDA Log

- An **EDA log** turns scattered plots into findings that others can check.
- Each entry records the question, the code and plot, and what you see, in one sentence with a number.
- It lists possible explanations, including data problems and confounders, and the next step.
- Report findings honestly as **hypotheses**; test them later on data not used to find them.

Example entry · Practice 5

Question: did the port affect survival?

Finding: 55.4% of Cherbourg passengers survived, against 33.7% from Southampton.

Explanations: 51% of Cherbourg passengers were in first class, against 20%.

Next: compare within class and sex; keep as a hypothesis.

⚠ **Common mistake:** writing an exploratory finding as a conclusion, such as “boarding at Cherbourg improved survival”.

Objective 02

Exercise: Which Chart?

Choose the main chart for each question about the Titanic data. Work in pairs for 3 minutes.

Question	Histogram	Scatter	Box by group	Bar / crosstab
Is fare skewed?				
Do older passengers pay more?				
Do fares differ between classes?				
Does survival differ by port and class?				
How many passengers came from each port?				

Why? Name the type of each variable first. Then add the summary that goes with the chart.

Key point: every chart needs a matching number: a median, a correlation, or a rate shown with its count.

Check Your Understanding

Hospital A has a higher overall death rate than hospital B. Yet for mild cases and for severe cases separately, A's death rate is lower than B's. Which explanation fits best?

A Hospital A is worse; the overall rate is the one that counts.

B A treats a much larger share of severe cases, so severity confounds the overall comparison.

C The figures must contain an error: an overall rate cannot exceed both group rates.

D Correlation is not causation, so no conclusion about the hospitals is possible.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Exploratory data analysis

Examining data with plots and summaries to find patterns and problems.

Confirmatory analysis

Testing a hypothesis fixed in advance with a planned procedure.

Univariate analysis

Describing one variable at a time with summaries and charts.

Bivariate analysis

Studying the relationship between two variables.

Pearson's r

The strength of a linear relationship, from −1 to +1.

Spearman's ρ

A rank correlation that captures any monotonic trend.

Confounder

A third variable that affects both the explanatory variable and the outcome.

Simpson's paradox

A trend in combined data that reverses or vanishes within groups.

Summary

In this lesson, you learned how to:

- **O1** Explain the purpose of EDA versus confirmatory analysis
- **O2** Decide the summary and chart for one or two variables
- **O3** Interpret a correlation, and why summaries are not enough
- **O4** Identify confounding and Simpson's paradox
- **O5** Design an EDA log entry that records a hypothesis

Key takeaway: explore to find good questions; plot every summary, compare within groups, and record findings as hypotheses.

Next

Practice 5 · Cleaning and EDA on a messy Titanic

Practice 5: Cleaning and EDA on a Messy Titanic

You receive a messy copy of a famous dataset, clean it step by step, and explore what it can tell you.

Goal

Find and fix quality problems, handle missing values, then explore survival and check a confounder.

Dataset

Titanic passengers: a messy copy of **903** rows and **16** columns, built from the clean 891-row file.

Notebook

The week 5 lab notebook, in Google Colab. No installation is needed.

Time

About 90 minutes: 60 in class, the rest at home before the week 6 session.

Parts A and B

Fix codes, types and duplicates; treat invalid ages; impute missing ages with a flag.

Part C

Build features, merge a port table safely, and explore survival by sex, class and port.

Do not edit the cell that creates the messy copy: everyone must clean the same problems.

Fixing Codes, Types and Duplicates

```python
# Python · Google Colab
df = messy.copy()
df["sex"] = df["sex"].str.strip().str.lower().replace(
    {"m": "male", "f": "female"})
df["fare"] = df["fare"].str.replace(" GBP", "", regex=False)\
    .str.replace(",", ".", regex=False).astype(float)
df = df.drop_duplicates(subset="passenger_id", keep="first")
```

1 Work on a copy; the raw data stays unchanged.

2–3 Seven spellings become male (579) and female (324).

4–5 Remove “GBP”, fix the comma, convert: no NaN left.

← **6 · Critical:** drop by the **ID**: 12 rows go, 891 remain. Never drop look-alike rows.

In practice

Start with messy.info() and value_counts(dropna=False) on sex, embark_town and age.

Invalid Ages and Missing Values

```python
# Python · Google Colab
bad_age = (df["age"] < 0) | (df["age"] > 100)
df.loc[bad_age, "age"] = np.nan
df.groupby("pclass")["age"].apply(lambda s: s.isna().mean())
df["age_missing"] = df["age"].isna().astype(int)
df["age"] = df.groupby(["pclass", "sex"])["age"].transform(
    lambda s: s.fillna(s.median()))
```

1–2 Four impossible ages become missing.

3 Share missing by class: 0.14, 0.06, 0.28.

← **4 · Critical:** create the flag **before** imputing, or the information is lost.

5–6 Fill each age with its class-and-sex median.

⚠ **Common mistake:** filling every missing age with the overall mean. It ignores that age is missing far more often in third class.

A Safe Merge and a Confounder

```
# Python · Google Colab
df = df.merge(ports, on="embarked", how="left",
              validate="many_to_one")
print(df.groupby("embark_town")["survived"].mean().round(3))
pd.crosstab(df["embark_town"], df["pclass"],
            normalize="index").round(2)
```

1–2 Add each port's country; the key must be unique in ports.

3 Survival by port: Cherbourg 0.554, Southampton 0.337.

← **4–5 · Your task:** Cherbourg is 51% first class. Now compute survival by port **within** each class.

In practice

Before the merge, build family_size, is_child and fare_per_person. Children under 16 survived 0.581 against 0.363.

What to Submit

Submit the completed notebook (.ipynb) on the course page. It must run from top to bottom without errors.

Part	What is marked	Points
Run	The notebook runs from top to bottom (Restart and run all)	1
A	Every quality problem listed with its dimension	2
A	Codes, types and duplicates fixed; look-alike rows explained	2
B	Invalid ages treated; mechanism argued; imputation with a flag	2
C	Features and a validated merge; survival within class computed	2
C	Three findings with numbers; the Cherbourg question answered	1
	Total	10

⚠ **Common mistake:** calling a pattern a cause. Cherbourg is a finding to explain, not a discovery about ports.

← **Due:** [date], before the week 6 session.

WEEK 5 · PAPERS TO READ

Papers to Read

A study of how analysts wrangle data, the paper that defined the mechanisms of missing data, and two classics on dirty data and exploration.

Required

Required Reading for Week 6

Read the listed sections before the week 6 class. One question in Quiz 6 may refer to these papers.

Wrangler: Interactive Visual Specification of Data Transformation Scripts
Kandel, Paepcke, Hellerstein & Heer · CHI 2011, ACM

Why wrangling is slow, and recording steps as a script. Read the introduction, the transformation language and the user study.

Inference and Missing Data
Rubin · Biometrika 63(3), 1976

The origin of MCAR and MAR. Read the summary and sections 1–2; skim the rest.

Bring a half-page note per required paper: the problem, the key idea, and its link to this week.

Optional

Optional Reading

A Taxonomy of Dirty Data

Kim, Choi, Hong, Kim & Lee · Data Mining and Knowledge Discovery 7(1), 2003

Dozens of kinds of dirty data, in a tree. Compare it with the six dimensions.

Exploratory Data Analysis, chapter 1

Tukey · Addison-Wesley, 1977 (book)

The founding text of EDA. Read the preface and chapter 1 for the attitude, not the techniques.

Optional readings are not examined. Choose one that relates to your project.

WEEK 5 · REQUIRED READING 1 · KANDEL ET AL. 2011

Required

Wrangler: Data Cleaning by Example and Preview

Kandel, S., Paepcke, A., Hellerstein, J., & Heer, J. (2011). Wrangler: Interactive visual specification of data transformation scripts. In Proceedings of CHI '11 (pp. 3363–3372). ACM. https://doi.org/10.1145/1978942.1979444

THE PROBLEM

Analysts spend much of their time reshaping data and fixing missing values, typos and odd formats. These steps are hard to specify (e.g. regular expressions), and manual edits in tools like Excel leave no reusable record.

KEY IDEAS

01 Select data in a table; the system suggests a ranked list of possible transforms

02 Each suggestion has a plain-language description and an in-place visual preview before you apply it

03 Every step is saved as a script that can be edited, annotated, exported as code and rerun on new data

Key term: Data wrangling — Reformatting data values or layout, fixing wrong or missing values, and combining sources so analysis can begin

Required

Cleaning Should Produce a Script, Not Just Clean Data

WHAT TO TAKE AWAY

1. In a 12-person study, median task time in Wrangler was over twice as fast as in Excel on all three tasks
2. Users rated previews and suggestions as more useful than editing transform parameters directly
3. Users without a mental model of the transform (e.g. reshaping) got stuck in both tools

THIS WEEK

Wrangler's operators (split, extract, fill, delete, fold/unfold) match pandas steps in 5-2. In the Titanic lab, keep cleaning in a notebook script, not hand edits, so it can be rerun.

LIMITATION

Small study: 12 users, tiny tables (at most 30 rows, 4 columns), three tasks. Gains on large, real data were expected, not measured.

For your half-page note: Why do the authors say the output of wrangling is a script, not only clean data? How does this apply to your Titanic notebook?

WEEK 5 · REQUIRED READING 2 · RUBIN 1976

Required

When Can We Ignore Why Data Are Missing

Rubin, D. B. (1976). Inference and missing data. Biometrika, 63(3), 581–592. https://doi.org/10.1093/biomet/63.3.581

THE PROBLEM

Analysts often analyse only the values they have and ignore the process that made other values missing. Rubin asks when this is valid, and shows that the answer depends on the type of inference used.

KEY IDEAS

01 Treat missingness as random data: an indicator per value with its own probability model

02 Missing at random (MAR): missingness may depend on observed values, not on the missing ones

03 Observed at random (OAR) adds that missingness does not depend on the observed values either

Key term: Ignorable missingness — When the missingness process can be left out of the model and inference from the observed data is still valid

Required

Ignoring Missingness Is an Assumption, Not a Default

WHAT TO TAKE AWAY

1. For likelihood and Bayesian inference, MAR plus distinct parameters lets us ignore the missingness process
2. For sampling-based (frequentist) inference, the data must be both MAR and observed at random
3. These ideas grew into the MCAR, MAR and MNAR labels used in later missing-data work

THIS WEEK

In 5-1 and the Titanic lab, ask why Age or Cabin is missing before you drop rows or fill values. dropna() and fillna() each assume a missingness mechanism.

LIMITATION

Mathematical and abstract. MAR cannot be tested from the observed data alone, so in practice it remains an assumption to argue for.

For your half-page note: For Titanic Age, is MCAR, MAR or MNAR most plausible? What would that mean for dropping rows versus filling the median?

Optional

A Taxonomy of Dirty Data

Kim, W., Choi, B.-J., Hong, E.-K., Kim, S.-K., & Lee, D. (2003). A taxonomy of dirty data. Data Mining and Knowledge Discovery, 7(1), 81–99. https://doi.org/10.1023/A:1021564703268

THE PROBLEM

Data mining and data warehouses depend on clean data, but there was no shared, complete list of the kinds of dirty data. Without one, it is hard to judge what a cleaning tool covers or misses.

KEY IDEAS

01 Classify dirty data in a hierarchy instead of a loose list of examples

02 Top split: missing data, wrong data, and data that is not wrong but still unusable

03 Link each type to how it can be prevented, detected or fixed, e.g. by database checks

Key term: Dirty data — Data that is missing, wrong, or correct but unusable as stored, e.g. inconsistent formats across sources

Optional

Not All Dirty Data Is Missing or Wrong

WHAT TO TAKE AWAY

1. Missing values are only one kind of dirty data; wrong values and unusable values need other methods
2. Some wrong data can be blocked by database constraints; much of it cannot and needs later checks
3. Correct values can still be unusable, e.g. different codes or spellings for the same thing

THIS WEEK

Use the taxonomy as a checklist for 5-1 and the lab: find missing values in messy Titanic, then wrong ones, then valid but inconsistent ones (e.g. mixed spellings).

LIMITATION

Built from a database and data-warehouse view; it lists problem types but gives no measured evidence of how often each occurs.

To think about: Give one Titanic example for each top-level class: missing, wrong, and not wrong but unusable. How would you detect each in pandas?

Optional

Look at the Data First: Stem-and-Leaf Displays

Tukey, J. W. (1977). Exploratory data analysis (Chapter 1: Scratching down numbers). Addison-Wesley.

THE PROBLEM

Analysts often jump to formal tests and models without first looking at the data. Lists of raw numbers hide the shape of a distribution, gaps and odd values.

KEY IDEAS

01 Exploratory analysis is detective work: look for clues before you confirm anything

02 Exploratory and confirmatory analysis are both needed; exploration comes first

03 Stem-and-leaf display: split numbers into stem and leaf to show shape and keep the values

Key term: Stem-and-leaf display — A text chart: leading digits form the stems, the next digit of each value is written as a leaf beside its stem

Optional

A Quick Display Can Show What a Table of Numbers Hides

WHAT TO TAKE AWAY

1. A stem-and-leaf shows a distribution's shape like a histogram while keeping the actual values
2. It makes clusters, gaps and unusual values easy to spot, and can be drawn by hand
3. Simple displays help decide which questions are worth testing formally later

THIS WEEK

This is the mindset of lesson 5-3. Before modelling survival, look at Titanic Age and Fare with describe(), histograms or a stem-and-leaf, and note gaps and odd values.

LIMITATION

Written for pencil-and-paper work on small data; stem-and-leaf displays become impractical for very large datasets.

To think about: Draw a stem-and-leaf of 20 Titanic ages. What does it show that the mean and standard deviation do not?

Discussion

Reading Questions for Discussion

- Wrangler records every transformation as a script. Why does that matter more than the speed of the tool?
- Rubin defines “missing at random”. Under which assumption did you impute the Titanic ages, and how could it fail?
- Pick one kind of dirty data from Kim et al. that the six dimensions of lesson 5-1 describe poorly.
- Tukey calls EDA detective work. What is the “judge and jury” in your project?

Prepare for class

- Read the assigned sections
- Write a half-page note per paper
- Bring one answer to each question
- Discuss in groups of three

Key point: a good answer takes a position and supports it with a passage from the paper or a result from Practice 5.

DS 501 · WEEK 5

This Week's Work

Practice 5 notebook

Finish the week 5 lab notebook, check that it runs from top to bottom, and submit the .ipynb file.

Due: [date]

Quiz 5

15 multiple-choice questions on lessons 5-1 to 5-3, plus one short applied question on a data quality report.

Due: [date]

Project milestone 2: data audit

Load your data, write the data dictionary and the quality report, argue the missing-data mechanism, and start an EDA log.

Due: [date]

Next week: Describing Data and Probability: robust summaries, the ideas of probability and sampling, and how to visualise distributions.